the annoying thing was that I was not to have at all the satisfaction of seeing myself vindicated. The universal acceptance of my thesis was ultimately brought about by considerations of an altogether different sort, unrelated to the arguments which I had adduced in support of it—namely, by the atomic theory, as represented by Ludwig Boltzmann.

Boltzmann succeeded in establishing, for a given gas in a given state, a function, *H*, which has the property that its value constantly decreases with time. It suffices, therefore, to identify the negative value of this *H* with entropy, to arrive at the principle of the increase of entropy. This discovery demonstrates, at the same time, irreversibility to be a characteristic of the processes occurring in a gas.

As events transpired, therefore, my claim concerning the fundamental difference between heat conduction and a purely mechanical process was victorious over the view previously entertained by outstanding authorities. Nevertheless, my contribution to the struggle was entirely superfluous, for even without it the outcome would have been the same.

Obviously, this battle, in which Boltzmann and Ostwald represented the opposing views, was fought rather heatedly, and produced also many a drastic effect, for the two antagonists were each other's equals in quick repartee and natural wit. After all that I have related, in this duel of minds I could play only the part of a second to Boltzmann—a second whose services were evidently not appreciated, not even noticed, by him. For Boltzmann knew very well that my viewpoint was basically different from his. He was especially annoyed by the fact that I was not only indifferent but to a certain extent even hostile to the atomic theory which was the foundation of his entire research. The reason was that at that time, I regarded the principle of the increase of entropy as no less immutably valid than the principle of the conservation of energy itself, whereas Boltzmann treated the former merely as a law of probabilities —in other words, as a principle that could admit of exceptions. The value of function *H* might also increase at times. Boltzmann did not go into this point in the deduction of his "*H*-Theorem," and a talented pupil of mine, E. Zermelo, noted emphatically this gap in a strict proof of the theorem. In fact, Boltzmann omitted in his deduction every mention of the indispensable presupposition of the validity of his theorem—namely, the assumption of molecular disorder. He must have simply taken it for granted. At any rate, he answered young Zermelo in a tone of biting sarcasm, which was meant partly for me, too, for Zermelo's paper had been published with my approval. This was how Boltzmann assumed that ill-tempered tone which he continued to exhibit toward me, on later occasions as well, both in his publications and in our personal correspondence; and it was only in the last years of his life, when I informed him of the atomistic foundation for my radiation law, that he assumed a friendlier attitude.